

VTVL Hopper Engineering Notebook

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1 Design Requirements

1.1 Mission Objective

Design a throttleable, simple liquid bipropellant engine (and every subsystem required for it to work) for a minimal VTVL hopper. The engine exists to support a possible vehicle in the future including subsystems such as avionics, landing gear, and test instrumentation. The theoretical flight objective for the full vehicle to be designed later is controlled liftoff, hover or near-hover control authority, and safe landing.

1.2 Requirement Classification

The requirements below are limited to items that must be true before the design is considered acceptable. Quantities that depend on vehicle mass convergence, tank selection, injector pressure drop, cooling margin, or throttle hardware are carried as sizing variables, not fixed requirements.

Item	Classification	Requirement / design rule
Budget	Hard	Prototype propulsion hardware target: $\leq \$5,000$ excluding propellant
Architecture	Hard	No turbopumps.
Cryogenics	Hard	No cryogenic propellant storage.
Hazard class and Legality	Hard	Nothing specifically potentially hazardous, corrosive, toxic, or illegal.
Material access	Hard	Primary wetted structural materials shall be readily obtainable aluminum alloy or stainless steel unless a later section gives a specific compatibility basis.
Fluid connectors	Hard	Use pressure-rated commercial connector families only: AN/MS, CGA, Swagelok-compatible compression fittings, or equivalent documented hardware.
Pressure safety	Hard	Pressure-wetted custom parts shall use burst factor of safety ≥ 4 against MEOP. Purchased cylinders, valves, regulators, and fittings shall be used only within documented manufacturer ratings.
Remote operation	Hard	All oxidizer and hot-fire operations shall be controllable from a safe remote location with positive propellant isolation before personnel approach.
Throttle demonstration	Test requirement	The engine shall demonstrate repeatable operation at full thrust and at lower commanded setpoints before flight. The final throttle ratio is set after vehicle mass is known.
Combustion stability	Test requirement	No sustained hard-start, chugging, buzz, or high-frequency instability is acceptable in qualification hot-fire data. Stability criteria are finalized in the injector and test sections.
TVC interface	Layout requirement	Reserve a defined thrust-frame or gimbal interface. Cooling jacket and plumbing geometry shall not block later TVC integration.

Table 1: Design requirements for the first prototype phase.

Quantity	Status	How it is used in this notebook
Nominal thrust	Sizing variable	Preliminary analyses may bracket 200-400 N, but the final value belongs in the engine sizing section after dry mass and propellant mass converge.
Full-thrust liftoff margin	Derived later	After m_0 is known, require $\frac{T_{max}}{m_0 g_0} \geq 1.2$ minimum. Higher margin is useful only if it does not force unnecessary system scale-up.
Minimum throttle	Derived later	After m_0 is known, require $T_{min} < m_0 g_0$ so the vehicle can command descent.
Burn duration	Design goal	30 s minimum useful demonstration; 60 s stretch target. Propellant mass is computed after thrust and $I_{sp,SL}$ are selected.
Chamber pressure	Design variable	Use 150 psia as the first static-fire analysis point and carry 200 psia as a performance-upgrade case if feed, cooling, and structure close.

Table 2: Sizing variables intentionally deferred to later notebook sections.

2 Propellant

2.1 Define the Problem

Select a storable oxidizer/fuel pair for a small pressure-fed VTVL hopper. The selected pair must support repeatable throttling, safe ground testing, affordable consumables, realistic tankage, and a credible path to injector and cooling design. The decision is not “maximize I_{sp} ”; it is “minimize total development risk while preserving enough performance for a minimal controlled hop.”

Categories of Analysis

- Cost
- Storage mass and volume
- Stability/Predictability
- Development risk
- Performance
- Complexity

2.2 Brainstorm Solutions

2.2.1 Hard Screens

These families fail a fixed constraint and are removed before detailed trade scoring.

Candidate	Reason for removal
LOX-based combinations	They violate the no-cryogenic first-article constraint. A LOX engine is a valid later project, not the fastest safe path to this hopper.
Hydrazine, MMH, UDMH, NTO	The toxicity and legal burden are incompatible with the intended small-team, low-budget test program.
RFNA/alcohol	Corrosive oxidizer handling and compatibility burden are not acceptable for the preferred aluminum/stainless build.
90%+ H_2O_2 / alcohol	High-test peroxide is a specialized oxidizer with severe decomposition and contamination hazards; sourcing and handling dominate the project.

2.2.2 Feasible Solutions

Candidate family	Why it is worth considering	Initial concerns to test
N_2O /alcohols	Self-pressurizing oxidizer, compact liquid storage, non-cryogenic, active small-VTVL precedent.	Two-phase injector flow, tank-temperature sensitivity, oxidizer pressure decay during burn.

GOX/alcohols	Single-phase oxidizer metering, stable regulator-fed pressure, easier analytical injector sizing.	Large high-pressure gas volume, regulator cost, oxygen cleaning, tank mass.
H ₂ O ₂ /alcohols	Dense storable oxidizer; 85% peroxide performance is close to N ₂ O/alcohol in the sea-level ambient comparison.	High-concentration peroxide sourcing, catalyst/ignition hardware, decomposition compatibility, hazardous transport.
N ₂ O or GOX with methanol	Methanol is cheap and historically common in rocket fuels.	Toxicity and no computed performance advantage over IPA/ethanol.
N ₂ O with gaseous fuels	Some spacecraft and hybrid literature exists; gaseous fuels can simplify atomization.	Poor fuel storage density and pressure-vessel mass for a 60 s ground hopper.
N ₂ O/RP-1 or kerosene	Dense fuel; common hydrocarbon fuel.	Coking/soot and regenerative cooling complexity are poor matches for a small reusable engine.

2.3 Screen the Candidate Set

The table below is generated from the companion RocketCEA project at $P_c = 150$ psia, $P_{amb} = 14.7$ psia, and a sea-level-matched nozzle. Values are shifting-equilibrium RocketCEA/NASA CEA outputs [1].

Candidate	$I_{sp,SL}$	O/F	ϵ	T_c K	c^* m/s	Screen result
GOX / IPA	230.1 s	1.65	2.43	3237	1762	Keep as GOX finalist. Best computed sea-level impulse, but GOX storage/hardware dominates.
GOX / Ethanol	225.6 s	1.55	2.45	3187	1727	Keep as GOX finalist. Slightly lower computed I_{sp} than GOX/IPA but stronger ignition/cooling evidence.
GOX / Methanol	221.3 s	1.20	2.46	3074	1693	Remove. No advantage over GOX/ethanol or GOX/IPA, with worse toxicity.
N ₂ O / IPA	207.0 s	5.10	2.37	3120	1590	Keep. Best N ₂ O/alcohol computed result and strongest VTVL heritage.
N ₂ O / Ethanol	204.8 s	4.65	2.38	3064	1573	Keep. Strong static-fire literature and better coolant conductivity.
85% H ₂ O ₂ / IPA	203.2 s	5.65	2.37	2553	1562	Near miss only. Dense oxidizer but sourcing/catalyst/decomposition burden is high.
N ₂ O / Methanol	203.0 s	3.50	2.38	2972	1559	Remove. No performance advantage and worse toxicity.
85% H ₂ O ₂ / Ethanol	201.1 s	5.00	2.36	2507	1547	Near miss only; same peroxide concerns.
70% H ₂ O ₂ / IPA	186.9 s	7.20	2.31	2184	1441	Remove. Low performance without solving peroxide access/catalyst risk.
70% H ₂ O ₂ / Ethanol	185.0 s	6.30	2.30	2139	1426	Remove. Low performance without solving peroxide access/catalyst risk.

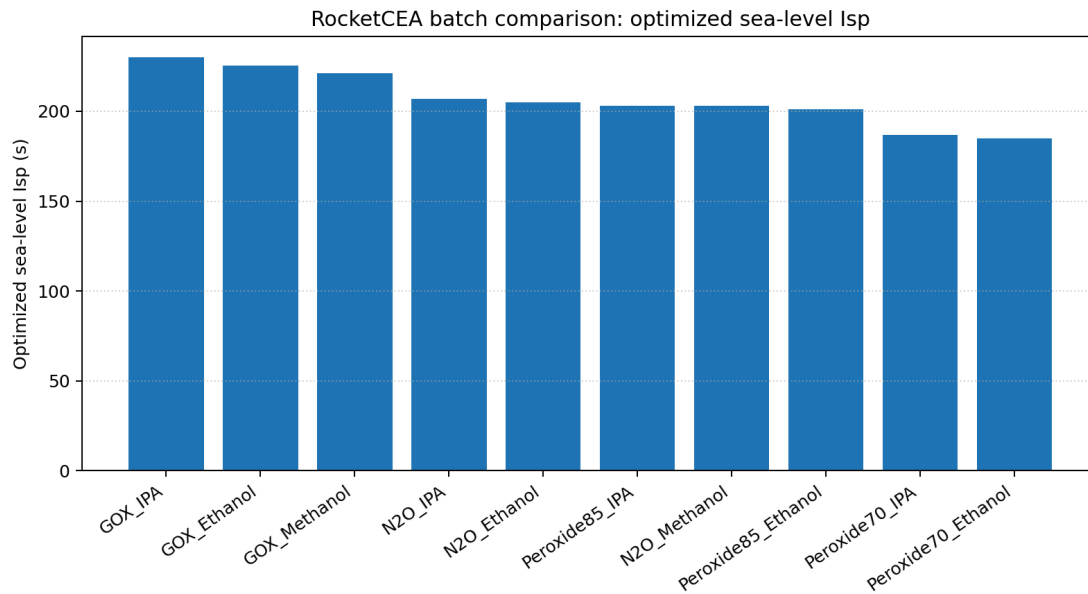


Figure 1: Generated RocketCEA comparison for optimized sea-level impulse at $P_c = 150$ psia, $P_{amb} = 14.7$ psia, shifting-equilibrium chemistry, and $P_e \approx P_{amb}$ nozzle sizing [1].

2.4 Quantitative Sourcing and System Sizing

Pricing below is a planning model, not a purchase instruction. Industrial gas prices vary by region, account status, deposit/rental policy, hazmat shipping, and cylinder exchange rules. The useful engineering comparison is the order of magnitude: GOX consumables can be cheap, but the storage/regulator system is expensive and heavy; N_2O consumables are more expensive per kilogram, but the liquid storage is much smaller and the tank is also the pressure source. I should replace these bands with local quotes before procurement, but public prices are enough to prevent the trade from pretending that propellant cost is the only cost.

Material	Practical acquisition route	Planning price	Design note
N_2O	Local welding/specialty gas, motorsport, food-service gas supplier	\$18–\$22/kg retail refill	Public motorsport refill pricing is \$8.49–\$9.79/lb. Local industrial quotes may be lower, but this is the sourced public band [2].
GOX	Welding oxygen cylinder exchange or industrial gas supplier	\$5–\$12/kg quote band	Public web pricing is often ZIP/login gated; a small oxygen refill listing and size-250 quote pages show that supplier quotes are required [3], [4].
99% IPA	Hardware, janitorial, lab, or chemical supplier	\$8–\$12/kg	Representative 99% IPA is \$23.50/gal before shipping/hazmat; 70% rubbing alcohol is not acceptable [5].
Anhydrous ethanol	Industrial/lab chemical supplier; denatured 200-proof acceptable if compatible	\$13–\$30/kg	Denatured 200-proof ethanol can be near \$40–\$57/gal, while tax-paid absolute ethanol can be much higher [6], [7].
70–85% H_2O_2	Specialty chemical supplier; high-concentration grades usually quote/hazmat	\$20–\$60+/kg placeholder	Public lower-concentration lab peroxide pricing is visible, but 70% is specialty/quote and not ordinary pool-supply grade [8], [9], [10].

Case	Thrust	Total prop.	Oxidizer	Fuel	Ox storage	Propellant cost
N_2O / IPA	200 N	5.91 kg	4.94 kg	0.97 kg	7.9 L	\$97–\$120
N_2O / IPA	400 N	11.82 kg	9.88 kg	1.94 kg	15.8 L	\$193–\$241
N_2O / Ethanol	200 N	5.97 kg	4.92 kg	1.06 kg	7.9 L	\$102–\$140

N ₂ O / Ethanol	400 N	11.95 kg	9.83 kg	2.11 kg	15.7 L	\$204–\$280
GOX / IPA	200 N	5.32 kg	3.31 kg	2.01 kg	17.5 L	\$33–\$64
GOX / IPA	400 N	10.64 kg	6.62 kg	4.01 kg	34.9 L	\$65–\$128
GOX / Ethanol	200 N	5.42 kg	3.30 kg	2.13 kg	17.4 L	\$44–\$103
GOX / Ethanol	400 N	10.85 kg	6.59 kg	4.25 kg	34.8 L	\$88–\$207

The GOX volume in the table assumes storage near 2200 psia and 293 K using the ideal-gas law; the N₂O volume assumes saturated-liquid storage near room temperature with a 20% ullage/design margin. These are comparison numbers, not final pressure-vessel dimensions. Candidate pressure vessels still need manufacturer ratings, valve details, fill rules, and compatibility checks [11].

The GOX number should be treated as a minimum gas-storage volume, because a regulated gas system must still finish the burn above the regulator delivery pressure and must include unusable residual gas. The 400 N, 60 s GOX cases already require about 35 L of 2200-psia oxygen storage before that residual allowance. A public size-250 oxygen cylinder is in the right volume class at roughly 43 L water volume, but it is a 108 lb steel cylinder before brackets and plumbing [4]. That can be acceptable as ground-support hardware, but it is a severe onboard mass penalty for a hopper.

Architecture	Public price anchors	Hardware allowance	Trade meaning
N ₂ O self-pressurizing liquid	10–20 lb nitrous bottles with high-flow valves list around \$315–\$465; race-ready 10–15 lb aluminum bottles list \$325–\$375 and tare around 15–19 lb [12], [13].	\$300–\$700 before common valves/sensors	Best budget/onboard storage match. It avoids the primary oxidizer regulator and pressurant bottle, but it buys two-phase injector, tank-temperature, and throttle calibration risk.
GOX regulated high-pressure gas	A new 250 cu ft CGA-540 oxygen cylinder lists around \$285 empty, and a representative high-pressure CGA-540 oxygen regulator lists around \$699 [14], [15].	\$650–\$1400 before common valves/sensors	Best analytical injector path. The gas is cheap, but oxygen-clean fittings, regulator capacity, residual pressure, and tank inert mass dominate the architecture.

This hardware table changes the survivor comparison more than the propellant-only table does. GOX has the highest CEA performance and the cleanest single-phase metering, but the first public cylinder-plus-regulator price anchors are already near \$1,000 before valves, instrumentation, brackets, and cleaning. N₂O is not analytically cleaner; it is simply much more compact and can use its own vapor pressure as the oxidizer feed source.

2.5 Peroxide and Methanol Near-Misses

85% H₂O₂/alcohol is not dismissed because of sea-level performance alone. In the baseline comparison it is close to N₂O/alcohol: 85% H₂O₂/IPA gives 203.2 s, only 3.8 s below N₂O/IPA [1]. It is removed because the supporting architecture is worse for this project: high-concentration peroxide sourcing is specialized, material compatibility is narrow, decomposition contamination can be violent, and practical engines often need a catalyst or dedicated ignition/decomposition hardware [10], [16].

70% H₂O₂ is not ordinary pool-supply grade in the sense needed here. Even if a 70% source is found, sea-level ambient-corrected performance is about 185–187 s at the baseline condition, while retaining the peroxide compatibility and handling burden [1].

Methanol combinations are also removed. Methanol does not outperform the matching IPA/ethanol cases in the baseline comparison, and the CDC/NIOSH pocket guide lists skin absorption as an exposure route, OSHA PEL of 200 ppm, and symptoms including visual disturbance and optic nerve damage [17]. That is a poor trade for no performance advantage.

2.6 Analyze Survivors

2.6.1 N₂O / IPA

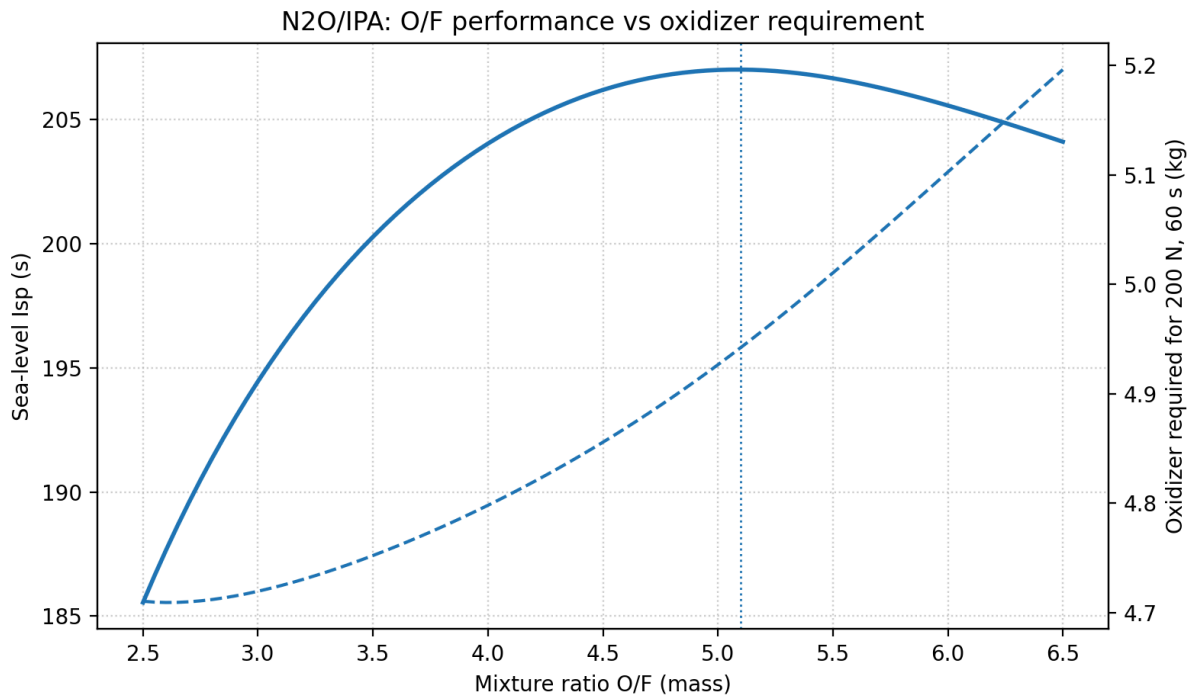


Figure 2: Individual O/F curve. The dashed curve in each panel is oxidizer mass required for a 200 N, 60 s burn.

The O/F curve also shows why “run fuel-rich to save oxidizer” is not a large win for this pair. At the 200 N, 60 s reference burn, moving from O/F = 5.10 to O/F = 4.50 cuts N₂O from 4.94 kg to 4.86 kg, but loses 0.8 s of computed sea-level I_{sp} and adds about 0.11 kg of IPA. At O/F = 4.00, the oxidizer savings are only about 0.14 kg while the penalty grows to about 3.0 s. I would not move far below the optimum only for oxidizer mass; I would shift O/F only if injector stability, cooling, or hot-fire data asks for it [1].

N₂O/IPA survives because it best matches the actual hopper mission. The sea-level ambient-corrected computed performance is $I_{sp,SL} = 207.0s$ at O/F = 5.10 and $\varepsilon = 2.37$ [1]. That is not high compared with vacuum-optimized numbers, but it is enough for a minimal hopper and avoids the hardware burden of high-pressure GOX storage.

The most important external evidence is direct VTVL heritage. Gruyère Space Program’s Colibri vehicle uses an N₂O/IPA bipropellant engine and is documented as a 2.45 m VTVL demonstrator with up to 1.25 kN thrust [18]. European Spaceflight reported the October 18, 2024 free flight to 105 m altitude, 30 m lateral translation, and 60 s duration [19]. That is the closest public analog to this project.

Decision Selected baseline. The main risk is not performance; it is injector/feed calibration for flashing N₂O. The advantage is compact oxidizer storage, no oxidizer regulator, cheap fuel, and direct small-VTVL precedent.

2.6.2 N₂O / Ethanol

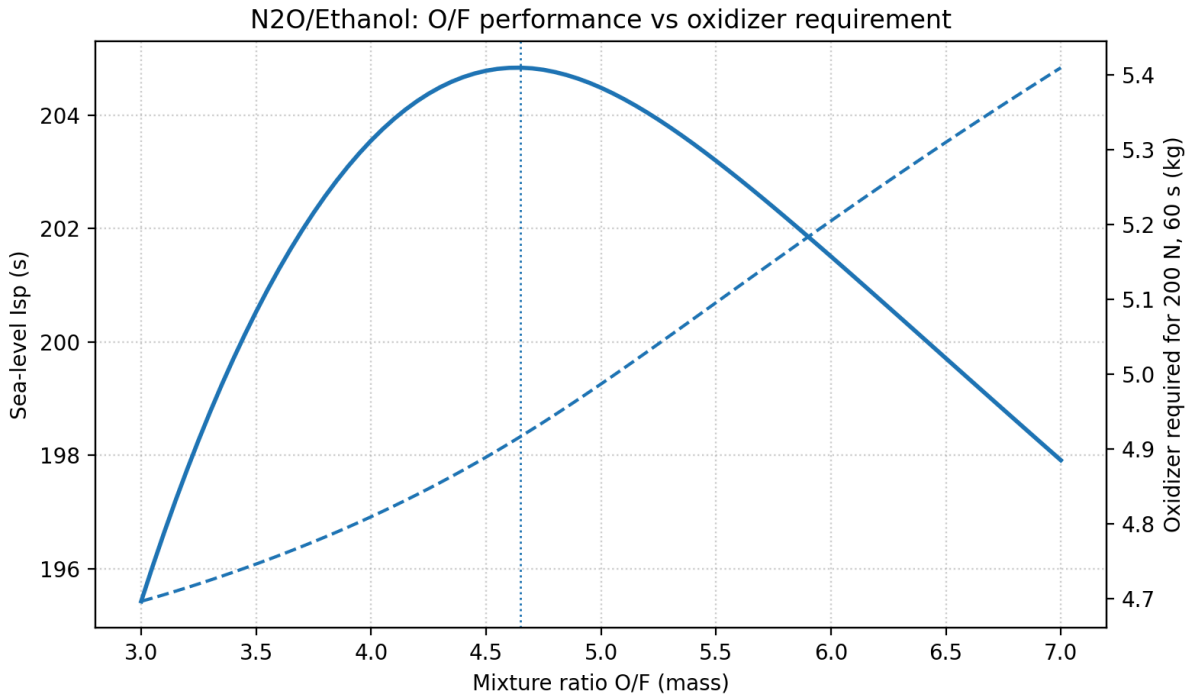


Figure 3: Individual O/F curve. The dashed curve in each panel is oxidizer mass required for a 200 N, 60 s burn.

For N₂O/ethanol the curve is broad near the optimum. O/F = 4.20 reduces N₂O from 4.92 kg to 4.84 kg and costs only 0.6 s, so a small fuel-rich bias is acceptable if it helps cooling or calibration. Going to O/F = 4.00 saves only about 0.11 kg oxidizer and already loses 1.3 s; O/F = 3.50 saves about 0.17 kg but loses 4.3 s. The mass benefit is too small to justify a large performance departure by itself [1].

N₂O/ethanol is the strongest backup because it has broader academic static-fire literature and better coolant thermal conductivity. The sea-level ambient-corrected computed performance is $I_{sp,SL} = 204.8$ s at O/F = 4.65 and $\varepsilon = 2.38$ [1]. The difference from N₂O/IPA is only 2.2 s in the baseline comparison, so it should not drive the decision.

Ethanol has slightly lower specific heat but higher thermal conductivity than IPA at room temperature; the coolant trade is therefore not one-sided. IPA has slightly better bulk heat capacity, while ethanol has better film-side conduction margin [20].

Decision Backup finalist. Use ethanol if the design is intentionally aligned with N₂O/ethanol literature or if regen-cooling margin dominates fuel procurement convenience.

2.7 N₂O Architecture Risk: Feed, Injector, and Environment

N₂O is stored as a saturated liquid at its own vapor pressure. Resonac lists high-purity N₂O vapor pressure as 5.24 MPa at 20°C and gives a critical temperature of 36.41°C with critical pressure 7.24 MPa [21]. This enables self-pressurization, but it also means feed pressure changes strongly with tank temperature.

▷ Las Vegas design-basis implication

A N₂O tank in direct summer sun can approach or exceed the 36.4°C critical temperature. The baseline control map should be built around measured tank temperature and pressure, and the pressure-vessel MEOP check should use the highest credible tank temperature, not just nominal room temperature. Testing in shade or morning conditions is not a convenience; it is part of keeping the feed model inside the calibrated regime.

When saturated liquid N₂O accelerates through valves and injector passages, pressure reduction can initiate flashing before the chamber. The standard single-phase incompressible equation is therefore only a bounding model:

$$\dot{m} = C_d A \sqrt{2\rho\Delta P}$$

N₂O injector design should bracket the flow with SPI and HEM and then use a non-equilibrium model such as Dyer/NHNE for the design estimate. Published work on self-pressurizing propellant flow and N₂O injectors supports this SPI/HEM/NHNE framing [22], [23], [24], [25].

Model	Assumption	Expected bias	Use in this project
SPI	Single-phase incompressible liquid.	Often overpredicts flashing N ₂ O flow.	Upper-bound geometry check only.
HEM	Two-phase thermodynamic equilibrium.	Often lower-bound for short injector residence times.	Lower-bound flow check.
Dyer / NHNE	Non-homogeneous, non-equilibrium interpolation between SPI and HEM using residence/bubble-growth times.	Best practical pre-test estimate for short rocket-injector passages.	Design-reference model, then calibrate by hot fire.
Empirical map	Valve command, tank state, and chamber response fitted from tests.	Most reliable inside tested range.	Required for throttle control.

The AEL N₂O/IPA throttle-control work is especially relevant because it reports closed-loop throttling on a small N₂O/IPA VTVL thruster, where flashing behavior made oxidizer pressure-drop behavior more thermodynamic than a simple hydraulic orifice prediction [26]. For this project, the practical implication is simple: N₂O injector and throttle design cannot be fully closed analytically. The engine needs calibration firings at multiple tank temperatures and throttle settings before flight. This is the main technical reason not to overscore N₂O just because its tanks are compact.

2.7.1 GOX / Ethanol

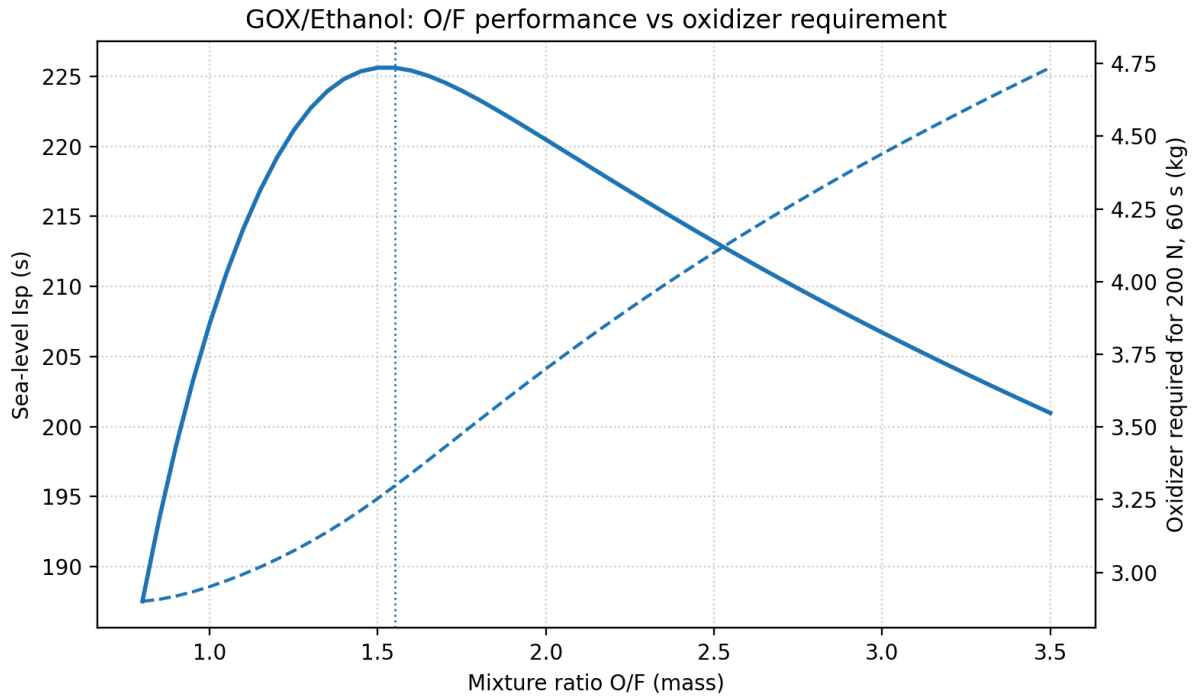


Figure 4: Individual O/F curve. The dashed curve in each panel is oxidizer mass required for a 200 N, 60 s burn.

GOX/ethanol has a more useful fuel-rich knob because oxygen storage is the GOX penalty. At the 200 N, 60 s reference burn, $O/F = 1.30$ saves about 0.19 kg GOX, or 5.8%, with a 2.9 s computed I_{sp} loss; this may be worth checking if tank volume is the limiting item. Pushing to $O/F = 1.20$ saves 0.25 kg GOX, or 7.6%, but loses 6.4 s and increases ethanol mass. The useful trade window is narrow, around $O/F = 1.30$ – 1.55 , not simply “as fuel-rich as possible” [1].

GOX/ethanol survives as the best alternative if the N_2O two-phase injector problem becomes unacceptable. The sea-level ambient-corrected computed performance is $I_{sp,SL} = 225.6s$ at $O/F = 1.55$ and $\varepsilon = 2.45$ [1]. It gives simpler single-phase oxidizer metering and stable regulator-fed pressure, but the system-level cost is high.

A NASA ignition characterization program tested GOX/ethanol at 150 psia and $O/F = 1.8$, making this the best-documented GOX/alcohol reference point near the baseline chamber pressure [27]. Oxygen-side hardware must be oxygen-cleaned and kept free of hydrocarbon contamination; the oxygen-cleaning and material-control burden is real, not paperwork [28].

Decision Engineering-risk alternate. Choose GOX/ethanol only if avoiding N_2O two-phase modeling is worth the larger gas storage, regulator cost, and oxygen-clean hardware process.

2.7.2 GOX / IPA

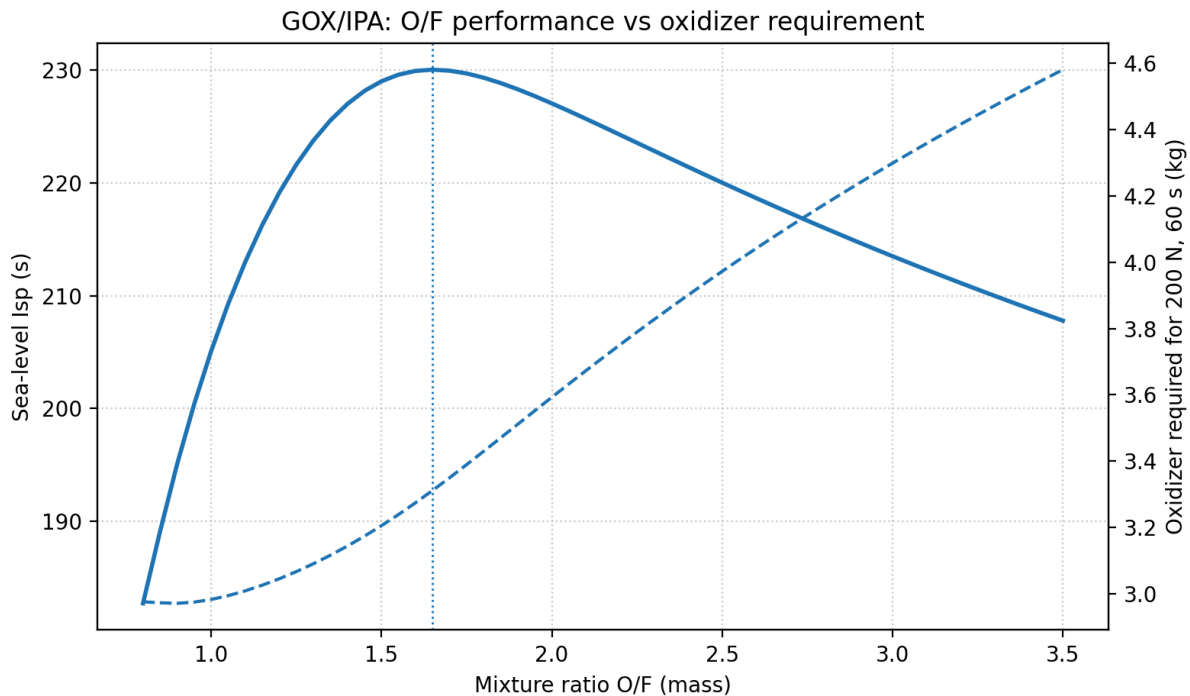


Figure 5: Individual O/F curve. The dashed curve in each panel is oxidizer mass required for a 200 N, 60 s burn.

GOX/IPA is less forgiving on the rich side. Dropping O/F from 1.65 to 1.40 saves about 0.17 kg GOX, or 5.1%, while losing 3.0 s, which could be tolerable if cylinder volume is marginal. Dropping to O/F = 1.20 saves only another 0.10 kg GOX beyond that but loses 10.9 s total, so the storage saving is not enough to justify the performance loss. If GOX/IPA is used, I would stay near O/F = 1.40–1.65 unless hot-fire stability points elsewhere [1].

GOX/IPA has the best sea-level ambient-corrected computed performance in this set: $I_{sp,SL} = 230.1s$ at O/F = 1.65 and $\varepsilon = 2.43$ [1]. The fuel is cheaper than anhydrous ethanol, but the fuel cost does not drive the GOX architecture. The expensive pieces are the oxidizer cylinder or custom gas vessel, high-flow oxygen regulator, CGA-540 adapters, oxygen-compatible valves, cleaning, and high-pressure fittings.

Decision Valid but not baseline. GOX/IPA is a credible GOX alternate. It is not selected over GOX/ethanol because the main GOX advantage is single-phase oxygen metering, and GOX/ethanol has stronger nearby ignition-validation evidence.

2.8 Pump Option Check

The hard rule is no turbopumps, not no pumps. I am still keeping the first-article survivor comparison pressure-fed because the pump does not remove the dominant oxidizer problems. For N_2O , self-pressurization is the benefit, and published injector work still treats pump/supercharge conditions as ways to set tank state or avoid cavitation, not as a way to eliminate flashing physics [23]. For GOX, a small electric liquid pump is not available because cryogenic LOX is screened out, and a gas compressor/feed pump would add oxygen-clean rotating hardware, battery power, controls, and heat without solving the large gas-storage requirement.

A small electric fuel pump remains allowed. It may become attractive later if the fuel tank or fuel pressurant mass dominates the vehicle, especially because IPA/ethanol are ordinary liquids compared with the oxidizers.

I am not adding it to the baseline yet because it adds a controller, battery/current margin, bypass or relief path, transient calibration, and a new single-point failure while leaving the oxidizer architecture unchanged.

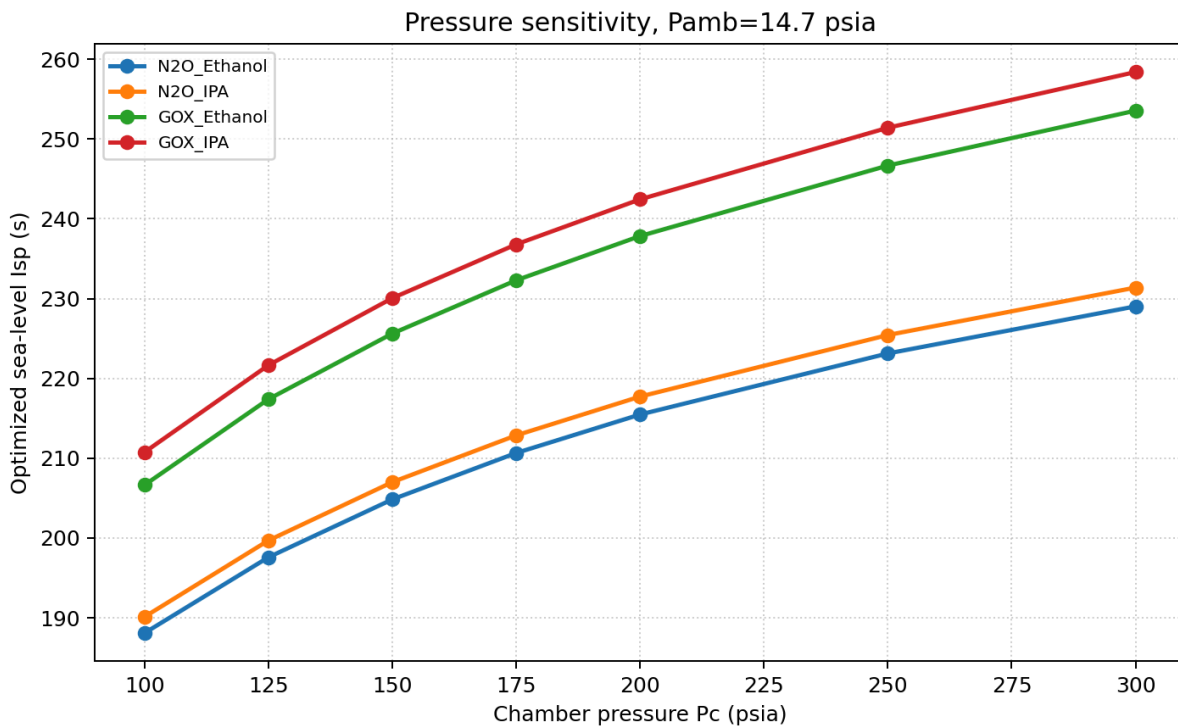


Figure 6: Pressure sensitivity for the four surviving alcohol finalists [1].

2.9 Select Best Solution

The high GOX performance and simple oxidizer metering are real. If this were only a static-fire injector development program, GOX/ethanol would be very hard to ignore. For an onboard budget hopper, though, the GOX bottle, regulator, oxygen-clean hardware, residual-pressure margin, and tank inert mass are the dominant penalties. The high peroxide density is also real, but sourcing and decomposition risk dominate.

Decision Baseline propellant selection: N_2O / IPA. N_2O /IPA is not selected because it is analytically easy; it is selected because it best fits the full hopper constraint set: compact self-pressurizing oxidizer storage, cheap and accessible fuel, and the most relevant small-VTVL flight precedent. The design must explicitly budget time for N_2O two-phase injector modeling and hot-fire calibration.

The best alternatives are conditional. **GOX/ethanol** is the cleanest technical alternate if avoiding two-phase N_2O behavior is worth the tank/regulator mass and cost. **N_2O /ethanol** remains the fuel backup if the project chooses to align with N_2O /ethanol literature or needs ethanol's higher thermal conductivity for cooling margin. **GOX/IPA** is credible and has the best computed I_{sp} , but it is not the reference GOX fuel because the fuel choice does not solve the GOX storage/regulator problem and GOX/ethanol has stronger nearby ignition-validation evidence.

3 Engine Sizing

4 Nozzle Design

5 Injector Design

6 Feed System Design

7 Cooling Analysis

8 Structural Analysis

9 CAD Design

10 Simulation

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